

Recent Advances in Pre-trained Neural Language Models

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Abstract

This paper presents a survey of recent advances in the field. We review key contributions, discuss methodological trends, and identify open challenges. Our analysis covers both foundational work and recent developments, highlighting directions for future research.

1. Introduction

Pre-trained language models have become the foundation of modern natural language processing. LSTM-based recurrent architectures [3] dominated sequence modelling until transformer-based models emerged. BERT [1] introduced masked language model pre-training, producing contextual representations that generalized across tasks. GPT-3 [2] demonstrated that scaling language models to hundreds of billions of parameters enables few-shot generalization without fine-tuning. Dropout regularization [5] remains widely used to prevent overfitting during pre-training.

2. Related Work

Vision-language models such as CLIP [4] align visual and textual representations using contrastive objectives over large web-crawled datasets. Hierarchical transformer designs have been proposed to handle long documents [6], while mixture-of-experts architectures offer scalable generation [7]. Quantum-inspired formulations have been applied to attention [8]. Pruning methods based on gradient analysis reduce model size [9], and curriculum strategies improve low-resource translation quality [10].

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